John Franklin Crenshaw

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Research: Using the high-redshift Universe to test dark energy, neutrino mass, and the growth of structure, turning the vast galaxy samples of Rubin, Roman, and future surveys into precision probes.

At a glance: 20 publications (5 first-author) plus 1 in review; 315 citations (h-index 10); $870,500 in fellowships and grants since 2024; 14 invited talks.

Professional Appointments

KIPAC Rubin Fellow, Stanford University & SLAC — Sep 2025 – present
Faculty Mentor: Pat Burchat

Education

University of Washington, MS, PhD in Physics — Sep 2019 – Jun 2025
Advisor: Andrew Connolly
Thesis: Towards High-redshift Cosmology with Lyman-break Galaxies Detected by LSST

Duke University, BS in Physics — Aug 2015 – May 2019
summa cum laude with highest distinction
Advisor: Kate Scholberg
Thesis: Sensitivity of the Helium and Lead Observatory to Core-Collapse Supernova Neutrino Bursts

Fellowships and Grants

KIPAC Rubin Fellowship ($368,000) — 2025 – 2029
Competitive independent postdoctoral fellowship.

NASA Euclid General Investigator Program, Science PI ($480,000) — 2025
Conceived the program and wrote the science case; fully funds a PhD student's salary.

KIPAC Workshop Grant, DESC DP2 Sprint Week ($16,000) — 2026

Dunlap Institute Workshop Grant ($6,500) — 2024

DOE Cosmic Frontier Grant ($360,000) — 2023
Led the proposal's cosmology component.

DOE SCGSR Fellowship ($10,400) — 2023

Rubin Observatory ISSC Ambassador ($4,500) — 2021 – 2022

DOE Scholar ($12,000) — 2021

Honors

Vera C. Rubin Observatory Builder Status — 2025
Awarded for essential contributions to Rubin's active optics system and commissioning.

LSST Dark Energy Science Collaboration (DESC) Builder Status — 2025
Awarded for key photo-z pipelines, founding the LBG Topical Team, and community leadership.

Research Highlights

Survey design: My LBG studies motivated a 40% increase in LSST u-band observing time.
Science case: My forecasts show LBGs break key degeneracies in LSST's low-redshift cosmology, improving its neutrino-mass constraint 5.6× and dark-energy figure of merit 2.6×.
First data: I am leading DESC's first LBG analyses, finding >9,000 candidates per deg^2 in Rubin DP2.
New science: Rubin's first CGM dust measurement, probing dust-rich halos around lower-mass galaxies.

Publications

As of November 2026, I have (co-)authored 20 publications with a total of 315 citations (h-index 10; full list on ADS). ⋆ marks my most significant papers.

First Author:

⋆ 5. *A First Measurement of Circumgalactic Dust Reddening from Only 4.6 deg^2 of the Rubin Observatory's Data Preview 1*
Crenshaw J. F., McQuinn M., Werk J. K. (2026) ApJL 1009 L13

⋆ 4. *Quantifying the Impact of LSST u-band Survey Strategy on Photometric Redshift Estimation and the Detection of Lyman-break Galaxies*
Crenshaw J. F., Leistedt B., Graham M. L., Payerne C., et al. (2025) ApJS 281 54

⋆ 3. *Probabilistic Forward Modeling of Galaxy Catalogs with Normalizing Flows*
Crenshaw J. F., Kalmbach J. B., Gagliano A., Yan Z., et al. (2024) AJ 168 80

2. *Using AI for Wave-front Estimation with the Rubin Observatory Active Optics System*
Crenshaw J. F., Connolly A. J., Meyers J. E., Kalmbach J. B., et al. (2024) AJ 167 86

1. *Learning Spectral Templates for Photometric Redshift Estimation from Broadband Photometry*
Crenshaw J. F., Connolly A. J. (2020) AJ 160 191

In Collaboration Review:

1. *Disentangling Shape and Intensity Information in Defocused Wavefront Estimation*
Crenshaw J. F., Megias Homar G., et al. (2026) *in Rubin Observatory internal review*

Co-Author with Major Contributions:

10. *Photometric redshift estimation for Rubin Observatory Data Preview 1 with Redshift Assessment Infrastructure Layers*
Zhang T., Charles E., **Crenshaw J. F.**, Schmidt S. J., et al. (2026) MNRAS 550 stag1012
My role: Built the reference sample and its Euclid cross-match.

9. *The Vera C. Rubin Observatory Data Preview 1*
Vera C. Rubin Observatory Team, including **Crenshaw J. F.** (2026) AJ 171 360
My role: Ran and optimized the active optics system to focus and collimate the telescope during the observations.

8. *Redshift Assessment Infrastructure Layers (RAIL): Rubin-era photometric redshift stress-testing and at-scale production*
RAIL Team, van den Busch J. L., Charles E., et al., including **Crenshaw J. F.** (2026) OJAp 9 58200
My role: Led development of RAIL's forward-modeling (creation) module: normalizing flows, photometric error models, and spectroscopic degradations.

7. *An Information-based Metric for Observing Strategy Optimization, Demonstrated in the Context of Photometric Redshifts for LSST*
Malz A. I., Lanusse F., **Crenshaw J. F.**, Scott B., et al. (2026) ApJS 282 69
My role: Created PZFlow, used in this work to estimate the information metric with conditional normalizing flows.

6. *Diagnosing the Effects of Spectroscopic Training Set Imperfection on Photometric Redshift Performance*
   Crafford A., Malz A. I., Zhang T., et al., including **Crenshaw J. F.** (2026) arXiv:2601.10797
   My role: Created PZFlow, used in this work to model photo-z biases.

5. *Impact of survey spatial variability on galaxy redshift distributions and the cosmological 3 × 2-point statistics for the Rubin Legacy Survey of Space and Time (LSST)*
   Hang Q., Joachimi B., Charles E., **Crenshaw J. F.**, et al. (2024) MNRAS 535 2970
   My role: Developed PhotErr and the RAIL forward-modeling tools used to propagate depth variations.

4. *The Active Optics System on the Vera C. Rubin Observatory: Optimal Control of Degeneracy among the Large Number of Degrees of Freedom*
   Megias Homar G., Kahn S. M., Meyers J. M., **Crenshaw J. F.**, et al. (2024) ApJ 974 108
   My role: Helped devise the SVD control strategy and its weighting by impact on image quality.

3. *Improving Photometric Redshift Estimates with Training Sample Augmentation*
   Moskowitz I., Gawiser E., **Crenshaw J. F.**, Andrews B. H., et al. (2024) ApJL 967 L6
   My role: Created PZFlow, used in this work to generate the augmented training galaxies.

2. *The simulated catalogue of optical transients and correlated hosts (SCOTCH)*
   Lokken M., Gagliano A., Narayan G., et al., including **Crenshaw J. F.** (2023) MNRAS 520 2887
   My role: Created PZFlow, used in this work to model correlated host-galaxy properties.

1. *The Sensitivity of GPz Estimates of Photo-z Posterior PDFs to Realistically Complex Training Set Imperfections*
   Stylianou N., Malz A. I., Hatfield P., **Crenshaw J. F.**, et al. (2022) PASP 134 044501
   My role: Created PZFlow, used in this work to model photo-z biases.

Other Co-Author Papers:
Includes papers on which my authorship reflects smaller contributions, data collection or calibration, or my Rubin and DESC Builder status, as well as white papers.

5. *Fast radio bursts in the era of the Vera C. Rubin Observatory's Legacy Survey of Space and Time*
   James C. W., Smith B., Dage K., et al., including **Crenshaw J. F.** (2026) PASA 43 e081

4. *NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1)*
   Chandler C. O., Bernardinelli P. H., Jurić M., et al., including **Crenshaw J. F.** (2026) ApJL 1001 L35

3. *A Path to an All-Sky Survey with Roman*
   Han J. J., Chiti A., Chen K., et al., including **Crenshaw J. F.** (2026) arXiv:2602.21280

2. *Photometric Redshifts in JWST Deep Fields: A Pixel-Based Alternative with DeepDISC*
   Merz G., Zhuang M., Li J., et al., including **Crenshaw J. F.** (2026) OJAp 9 56099

1. *Opportunities in AI/ML for the Rubin LSST Dark Energy Science Collaboration*
   LSST Dark Energy Science Collaboration, including **Crenshaw J. F.** (2026) arXiv:2601.14235

Presentations

Invited Talks:

California Institute of Technology Seminar	*Pasadena, CA*; Oct 2026
Kavli Symposium Plenary	*Cambridge, UK*; Sep 2026
Rubin Observatory Community Workshop Plenary	*Palo Alto, CA*; Jul 2026
HSC Medium-band Survey Science Community Workshop	*Nagoya, Japan*; Jun 2026
University of California, Santa Cruz Seminar	*Santa Cruz, CA*; Feb 2026
Pontificia Universidad Católica de Valparaíso Colloquium	*Valparaíso, Chile*; Oct 2025
Berkeley Center for Cosmological Physics Seminar	*Berkeley, CA*; Sep 2025
DESI-II Working Group	*Virtual*; Feb 2025
DES-DESC Special Session, AAS 245th Meeting	*National Harbor, MD*; Jan 2025
Cosmopalooza Plenary	*Virtual*; Oct 2023
University of Chile Colloquium	*Santiago, Chile*; Mar 2023
AAS Astronomers Turned Data Scientists Meeting Plenary	*Virtual*; Mar 2022
DESC Winter Meeting Plenary	*Virtual*; Feb 2022
KIPAC Seminar	*Virtual*; Sep 2020

Recent Contributed Talks and Posters:

Talk, Rubin Early Results Special Session, AAS 247th Meeting	*Phoenix, AZ*; Jan 2026
Poster, New Synergies in Multi-Probe Cosmology	*Santa Barbara, CA*; Feb 2026
Poster, AAS 247th Meeting	*Phoenix, AZ*; Jan 2026
Poster, Adaptive Optics for Extremely Large Telescopes 8	*Viña del Mar, Chile*; Oct 2025

Mentoring

Outcomes: two first-author papers in preparation, a machine-learning model deployed in Rubin operations, and conference presentations by every PhD mentee.

PhD Students:

Ben Sherwin, Stanford University (advisor: Josh Frieman) 2025 – present
LBG photo-z requirements for LSST Year 1 cosmology.

Leonid Sajkov, Stanford University (advisor: Pat Burchat) 2026 – present
Deep-field simulations for LBG redshift calibration.

Hanyue Wang, Stanford University (advisor: Risa Wechsler) 2026 – present
Small-scale LBG clustering in Rubin DP2 to constrain galaxy bias.

Linda Jin, University of Washington (advisor: Andrew Connolly) 2025 – present
Deep-learning spectral models of LBGs; funded by my NASA Euclid grant.

Hurum Tohfa, University of Washington (advisor: Andrew Connolly) 2024 – present
Deep-learning wavefront estimation for Rubin's active optics; model deployed in on-sky operations.

Undergraduate Students:

Dominik Riemann, University of Washington (advisor: Andrew Connolly) 2022 – 2024
Deep-learning wavefront estimation for Rubin's active optics.

Teaching and Curriculum Development

Course Designer & Instructor, Reading Courses *University of Washington*; 2020 – 2022
Designed curricula for and taught *Tensions in ΛCDM Cosmology* and *Gravitational Lensing*.

Lecturer, DESC Dark Energy School — *Boston University*; 2026
Designed an interactive lesson on AI tools for cosmology research.

Guest Lecturer — *Stanford University*; 2026
Gave two lectures in a graduate course on the Rubin Observatory and LSST.

Guest Lecturer — *University of Washington*; 2021 – 2026
Lectured in graduate cosmology (2021, 2026) and extragalactic astronomy (2025).

Postdoc Mentoring Certificate (in progress) — *Stanford University*; 2025 – present
Includes observing active-learning techniques in upper-level physics courses.

Member, Reading Course Leadership Committee — *University of Washington*; 2022

Teaching Assistant — *Duke University*; 2016 – 2019

Outreach and Public Speaking

Organizing:

Co-organizer, Astro on Tap San Francisco — *San Francisco, CA*; 2025 – present
Recruit speakers for, advertise, and host monthly public talks for audiences of about 250.

Organizer & Pedagogical Simulation Lead, STEM Pals — *Virtual*; 2020 – 2021
Helped launch the program; designed interactive simulations for high school students.

Public Talks:

Lectures on Tap: *Physics of the Cosmic Speed Limit* — *San Francisco, CA*; Nov 2026

Lectures on Tap: *Mapping the Dark Universe with Rubin* — *San Francisco, CA*; May & Aug 2026

De Anza College Planetarium: *The Rubin Observatory* — *Cupertino, CA*; Apr 2026

Astro on Tap: *Objects in Mirror Are Farther Than They Appear* — *San Francisco, CA*; Oct 2025

Astro on Tap: *Dark Matter Murder Mystery* — *Seattle, WA*; Mar 2025

Emerald City Comic Con: Rubin Observatory Panel — *Seattle, WA*; Mar 2025

Astro on Tap: *Dark Energy in the Era of DESI* — *Seattle, WA*; May 2024

Astro on Tap: *Before the Big Bang* — *Seattle, WA*; Apr 2023

Earlier Outreach (2018 – 2022): a middle school class in Prague, a public lecture and star parties at Duke University, and two panels for LGBTQ+ students in STEM.

Leadership and Service

Founder & Leader, DESC Lyman-break Galaxy Topical Team — 2024 – present
Lead a 14-member team making LBGs a precision cosmological probe, including its first Rubin analyses.

Co-chair, DESC Equity, Diversity, and Inclusion Committee — 2023 – present
Advise the Collaboration Council and management; co-designed and ran the climate survey; created comprehensive safety resources; improved training of meeting contact persons.

Organizer, DESC DP2 Sprint Week at SLAC — 2026
Week-long DESC analysis sprint on Rubin Data Preview 2 for 50 participants.

Co-organizer, Workshop on Cosmology with LBGs, Dunlap Institute — 2025
Brought together 30 Rubin and DESI researchers on LBG selection, calibration, and cosmology.

Member, Rubin 2036 Workshop Science & Local Organizing Committees — 2026

Member, Rubin Community Workshop Science Organizing Committee — 2023 – 2024

Member, DESC Collaboration Meeting Science Organizing Committee — 2022 – 2023

Participant, HEP Funding Advocacy Trip to Washington, D.C. — 2026

Referee for ApJ, JCAP, OJAp, and the ICML ML4Astro Workshop 2025 – present

Officer, Physicists for Inclusion and Equity 2020 – 2021

Observing and Commissioning

Active Optics Commissioning, Rubin Observatory 2023 – 2025
Led much of the on-sky commissioning of the active optics system over six months on the summit; focused and collimated the telescope for Data Previews 1 and 2.

Observer, IBIS Medium-band Survey, Blanco/DECam 2026
Observed five nights of imaging for DESI Run 2 spectroscopic targeting of Lyman-alpha emitters.

Lead Proposer, Subaru HSC-Niji Medium-band Program 2026
Semester S27A proposal, under review, to calibrate LBG redshift distributions in COSMOS.

Software

`PZFlow`: Creator and lead developer of a normalizing-flow package for forward-modeling galaxy catalogs; its paper, shared with PhotErr, has 38 citations in two years.

`PhotErr`: Creator and lead developer of a photometric-error model for Rubin, Euclid, and Roman; used in Roman simulations, LSST strategy studies, and DESC forecasts.

`lbg_tools`: Creator of a package modeling LBG populations and selection yields versus survey depth; underpins the u-band study, LBG forecasts, and photo-z requirements.

`RAIL`: Lead developer of forward modeling in DESC's photo-z framework, which generates realistic catalogs and systematic errors for validating photo-z methods.

`ts-wep`: Core developer of the wavefront pipeline driving Rubin's active optics; redesigned its architecture to integrate deep-learning and forward-modeling estimators.